

PROBLEM 15.56

Knowing that crank AB has a constant angular velocity of 160 rpm counterclockwise, determine the angular velocity of rod BD and the velocity of roller D when $\theta = 40^\circ$.

SOLUTION

Crank AB :

$$\omega_{AB} = 160 \text{ rpm} = 16.255 \text{ rad/s}$$

$$AB = 3 \text{ in.}$$

$$v_B = (AB)\omega_{AB}$$

$$= (3 \text{ in.})(16.255 \text{ rad/s})$$

$$v_B = 50.27 \text{ in./s } \nearrow 30^\circ$$

Rod BD :

$$\beta = \sin^{-1} \frac{3.402}{19} = 19.89^\circ$$

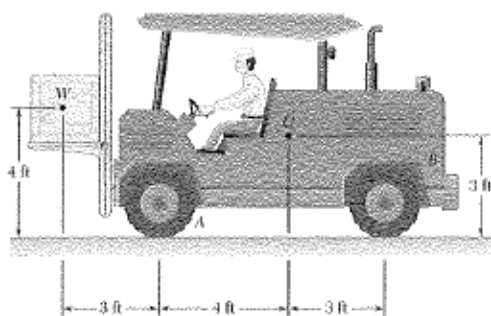
Law of sines:

$$\frac{v_D}{\sin 40^\circ} = \frac{v_B}{\sin 30^\circ} = \frac{50.27 \text{ in./s}}{\sin 19.89^\circ}$$

$$v_D = 34.44 \text{ in./s}$$

$$v_{BD} = (10 \text{ in.})\omega_{BD} = 26.73 \text{ in./s}$$

$$\omega_{BD} = 2.67 \text{ rad/s } \searrow$$



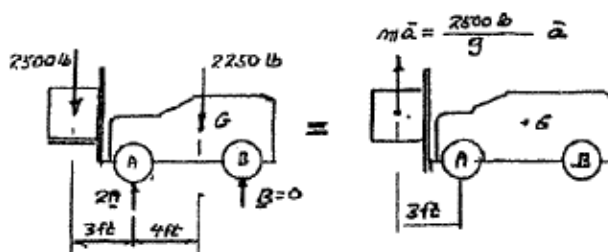
PROBLEM 16.9

The forklift truck shown weighs 2250 lb and is used to lift a crate of weight $W = 2500$ lb. Knowing that the truck is at rest, determine (a) the upward acceleration of the crate for which the reactions at the rear wheels B are zero, (b) the corresponding reaction at each of the front wheels A .

SOLUTION

(a) Acceleration of crate for

$$B = 0.$$



$$+\circlearrowleft \Sigma M_A = \Sigma (M_A)_{\text{eff}}: (2500 \text{ lb})(3 \text{ ft}) - (2250 \text{ lb})(4 \text{ ft}) = -m\bar{a}(3 \text{ ft})$$

$$7500 - 9000 = \frac{2500}{g}\bar{a}(3)$$

$$\bar{a} = \frac{1}{5}g = \frac{1}{5}(32.2 \text{ ft/s}^2)$$

$$\bar{a} = 6.44 \text{ ft/s}^2 \uparrow \blacktriangleleft$$

(b) Reaction at A:

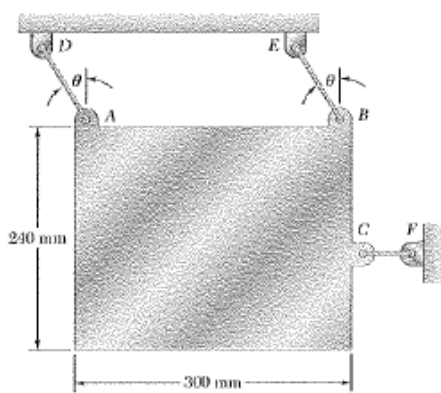
$$+\uparrow \Sigma F_y = \Sigma (F_y)_{\text{eff}}: 2A - 2500 \text{ lb} - 2250 \text{ lb} = m\bar{a}$$

$$2A - 4750 \text{ lb} = \frac{2500 \text{ lb}}{g} \left(\frac{g}{5} \right)$$

$$2A = 5250 \text{ lb}$$

For one wheel:

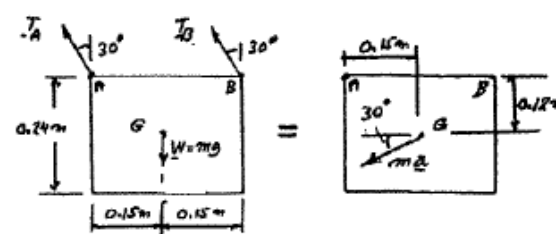
$$A = 2625 \text{ lb} \uparrow \blacktriangleleft$$



PROBLEM 16.14

A uniform rectangular plate has a mass of 5 kg and is held in position by three ropes as shown. Knowing that $\theta = 30^\circ$, determine, immediately after rope CF has been cut, (a) the acceleration of the plate, (b) the tension in ropes AD and BE .

SOLUTION



(a) Acceleration $+\nearrow 30^\circ \Sigma F = \Sigma F_{\text{eff}}: mg \sin 30^\circ = m\bar{a}$

$$\bar{a} = 0.5g = 4.905 \text{ m/s}^2 \qquad \bar{a} = 4.91 \text{ m/s}^2 \nearrow 30^\circ \blacktriangleleft$$

(b) Tension in ropes

$+\curvearrowright \Sigma M_A = \Sigma (M_A)_{\text{eff}}: (T_B \cos 30^\circ)(0.3 \text{ m}) - mg(0.15 \text{ m}) = -m\bar{a}(\cos 30^\circ)(0.12 \text{ m}) - m\bar{a}(\sin 30^\circ)(0.15 \text{ m})$

$$0.2598T_B - (5 \text{ kg})(9.81 \text{ m/s}^2)(0.15 \text{ m}) = -(5 \text{ kg})(4.905 \text{ m/s}^2)(0.1039 + 0.07)$$

$$0.2598T_B - 7.3575 = 4.388$$

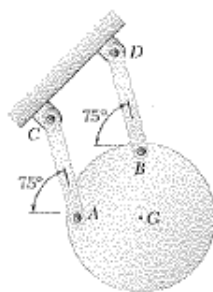
$$T_B = +11.43 \text{ N} \qquad T_{BE} = 11.43 \text{ N} \blacktriangleleft$$

$+\searrow 10^\circ \Sigma F = \Sigma F_{\text{eff}}: T_A + 11.43 \text{ N} - mg \cos 30^\circ = 0$

$$T_A + 11.43 \text{ N} - (5 \text{ kg})(9.81) \cos 30^\circ = 0$$

$$T_A + 11.43 \text{ N} - 42.48 \text{ N} = 0$$

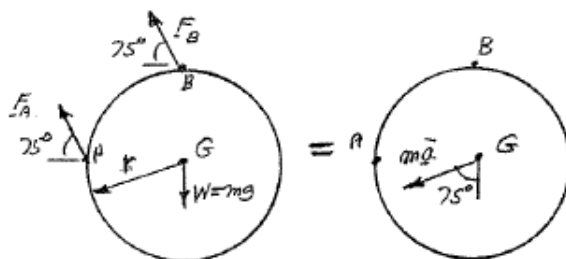
$$T_A = 31.04 \text{ N} \qquad T_{AD} = 31.0 \text{ N} \blacktriangleleft$$



PROBLEM 16.16

A uniform circular plate of mass 3 kg is attached to two links AC and BD of the same length. Knowing that the plate is released from rest in the position shown, in which lines joining G to A and B are, respectively, horizontal and vertical, determine (a) the acceleration of the plate, (b) the tension in each link.

SOLUTION



(a) Acceleration.

$$+\nearrow \Sigma F = \Sigma F_{\text{eff}}: \quad mg \cos 75^\circ = m\bar{a}$$

$$\bar{a} = g \cos 75^\circ = 2.5390 \text{ m/s}^2 \quad \bar{a} = 2.54 \text{ m/s}^2 \nearrow 15^\circ \blacktriangleleft$$

(b) Tension in each link.

$$+\curvearrowright \Sigma M_B = \Sigma (M_B)_{\text{eff}}: \quad (F_A \cos 75^\circ)r + (F_A \sin 75^\circ)r = (m\bar{a} \sin 75^\circ)r$$

$$F_A (\cos 75^\circ + \sin 75^\circ) = (3 \text{ kg})(2.539 \text{ m/s}^2) \sin 75^\circ$$

$$F_A = 6.01 \quad F_A = 6.01 \text{ N Tension} \blacktriangleleft$$

$$+\searrow 75^\circ \Sigma F = \Sigma F_{\text{eff}}: \quad F_A + F_B - mg \sin 75^\circ = 0$$

$$6.009 \text{ lb} + F_B - (3 \text{ kg})(9.81 \text{ m/s}^2) \sin 75^\circ = 0$$

$$F_B = 22.42 \text{ N} \quad F_{BD} = 22.4 \text{ N Tension} \blacktriangleleft$$

15.66 En la posición mostrada, la barra DE tiene una velocidad angular constante de 10 rad/s en el sentido de las manecillas del reloj. Si $h = 500 \text{ mm}$, determine $a)$ la velocidad angular de la barra FBD , $b)$ la velocidad del punto F .

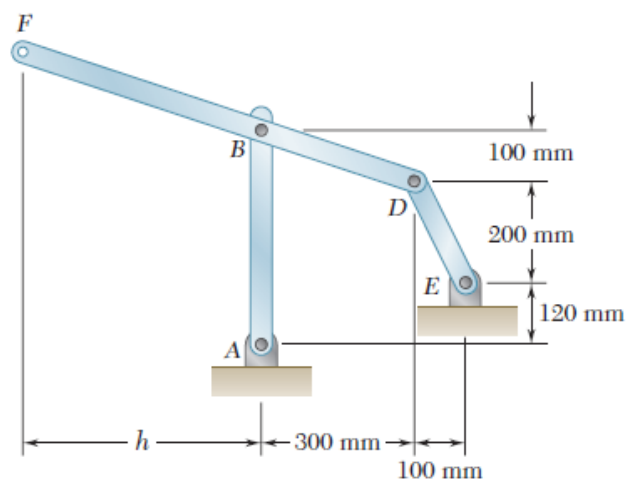
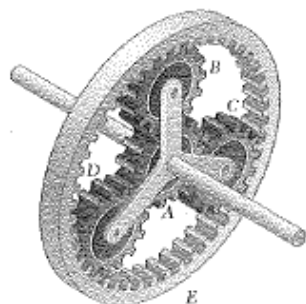


Figura P15.66 y P15.67

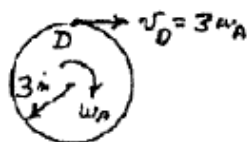


PROBLEM 15.48

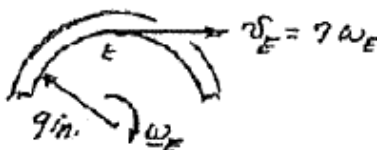
In the planetary gear system shown, the radius of gears A , B , C , and D is 3 in. and the radius of the outer gear E is 9 in. Knowing that gear E has an angular velocity of 120 rpm clockwise and that the central gear has an angular velocity of 150 rpm clockwise, determine (a) the angular velocity of each planetary gear, (b) the angular velocity of the spider connecting the planetary gears.

SOLUTION

Gear A



Outer gear E



Planetary gear.

Components $\pm \rightarrow$:

$$v_B = v_D + 6\omega_B$$

$$9\omega_E = 3\omega_A + 6\omega_B$$

$$9(120 \text{ rpm}) = 3(150 \text{ rpm}) + 6\omega_B$$

(a)

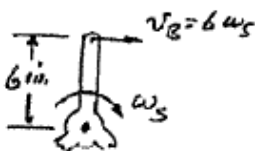
$$\omega_B = 105 \text{ rpm}$$

$$\omega_B = 105 \text{ rpm} \curvearrowright \blacktriangleleft$$

Components $\pm \rightarrow$:

$$\begin{aligned} v_B &= v_D + 3\omega_B - 3\omega_A + 3\omega_B \\ &= 3(\omega_A + \omega_B) \end{aligned}$$

Spider



$$v_B = 6\omega_S$$

$$3(\omega_A + \omega_B) = 6\omega_S$$

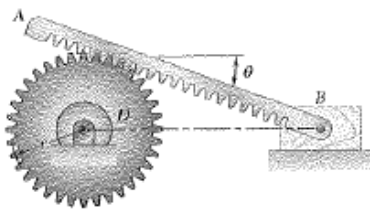
$$\omega_S = \frac{1}{2}(\omega_A + \omega_B)$$

$$= \frac{1}{2}(150 \text{ rpm} + 105 \text{ rpm})$$

(b)

$$\omega_S = 127.5 \text{ rpm}$$

$$\omega_S = 127.5 \text{ rpm} \curvearrowright \blacktriangleleft$$



PROBLEM 15.59

A straight rack rests on a gear of radius r and is attached to a block B as shown. Denoting by ω_D the clockwise angular velocity of gear D and by θ the angle formed by the rack and the horizontal, derive expressions for the velocity of block B and the angular velocity of the rack in terms of r , θ , and ω_D .

SOLUTION

Gear D : Rotation about D . Tooth E is in contact with rack AB .

$$v_E = r\omega_D \swarrow \theta$$

Rack AB :

$$l_{EB} = \frac{r}{\tan \theta}$$

Plane motion = Translation with E + Rotation about E .



$$v_B = v_E + v_{BE} \quad [v_B \rightarrow] = [v_E \swarrow \theta] + [v_{BE} \nearrow \theta]$$

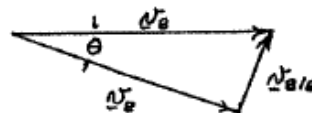
Draw velocity vector diagram.

$$v_B = \frac{v_E}{\cos \theta} = \frac{r\omega_D}{\cos \theta}$$

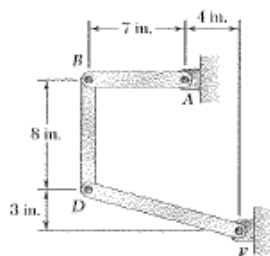
$$v_B = \frac{r\omega_D}{\cos \theta} \rightarrow$$

$$v_{BE} = v_E \tan \theta = r\omega_D \tan \theta$$

$$\begin{aligned} \omega_{AB} &= \frac{v_{BE}}{l_{EB}} \\ &= \frac{r\omega_D \tan \theta}{\frac{r}{\tan \theta}} \\ &= \omega_D \tan^2 \theta \end{aligned}$$



$$\omega_{AB} = \omega_D \tan^2 \theta \curvearrowright$$



PROBLEM 15.63

In the position shown, bar AB has an angular velocity of 4 rad/s clockwise. Determine the angular velocity of bars BD and DE .

SOLUTION

Bar AB : (Rotation about A) $\omega_{AB} = 4 \text{ rad/s} \Rightarrow -(4 \text{ rad/s})\mathbf{k}$

$$\mathbf{r}_{B/A} = -(7 \text{ in.})\mathbf{i} \quad \mathbf{v}_B = \omega_{AB} \times \mathbf{r}_{B/A} = (-4\mathbf{k}) \times (-7\mathbf{i})$$

$$\mathbf{v}_B = (28 \text{ in./s})\mathbf{j}$$

Bar BD : (Plane motion = Translation with B + Rotation about B .)

$$\omega_{BD} = \omega_{BD}\mathbf{k} \quad \mathbf{r}_{D/B} = -(8 \text{ in.})\mathbf{j}$$

$$\mathbf{v}_D = \mathbf{v}_B + \omega_{BD} \times \mathbf{r}_{D/B} = 28\mathbf{j} + (\omega_{BD}\mathbf{k}) \times (-8\mathbf{j})$$

$$\mathbf{v}_D = 28\mathbf{j} + 8\omega_{BD}\mathbf{i}$$

Bar DE : (Rotation about E)

$$\omega_{DE} = \omega_{DE}\mathbf{k}$$

$$\mathbf{r}_{D/E} = -(11 \text{ in.})\mathbf{i} + (3 \text{ in.})\mathbf{j}$$

$$\mathbf{v}_D = \omega_{DE} \times \mathbf{r}_{D/E} = (\omega_{DE}\mathbf{k}) \times (-11\mathbf{i} + 3\mathbf{j})$$

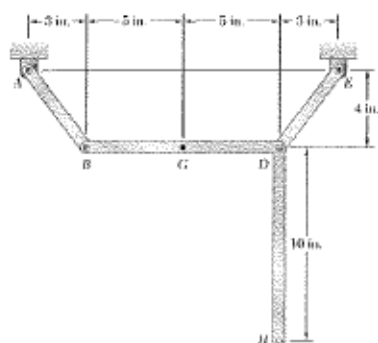
$$\mathbf{v}_D = -11\omega_{DE}\mathbf{j} - 3\omega_{DE}\mathbf{i}$$

Equating components of the two expressions for $\mathbf{v}_D$,

$$\mathbf{j}: 28 = -11\omega_{DE} \quad \omega_{DE} = -2.5455 \text{ rad/s} \quad \omega_{DE} = 2.55 \text{ rad/s} \quad \blacktriangleleft$$

$$\mathbf{i}: 8\omega_{BD} = -3\omega_{DE} \quad \omega_{BD} = -\frac{3}{8}\omega_{DE}$$

$$\omega_{BD} = -\left(\frac{3}{8}\right)(-2.5455) = 0.95455 \text{ rad/s} \quad \omega_{BD} = 0.955 \text{ rad/s} \quad \blacktriangleleft$$

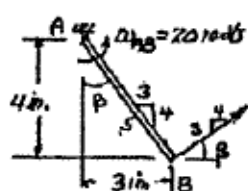


PROBLEM 15.68

In the position shown, bar AB has zero angular acceleration and an angular velocity of 20 rad/s counterclockwise. Determine (a) the angular velocity of member BDH , (b) the velocity of Point G .

SOLUTION

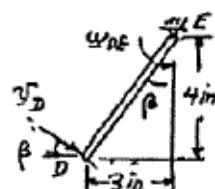
Bar AB :



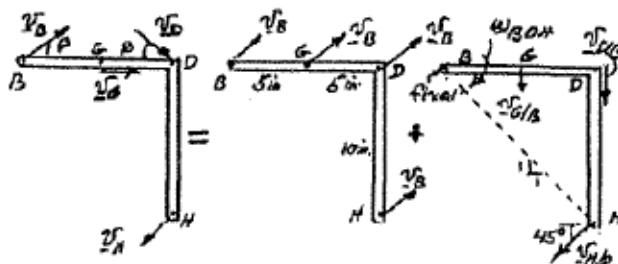
$$AB = 5 \text{ in.} \quad \beta = \tan^{-1} \frac{3}{4} = 36.87^\circ$$

$$v_B = (5 \text{ in.})(20 \text{ rad/s}) = 100 \text{ in./s}$$

Bar DE :



Member BDH :



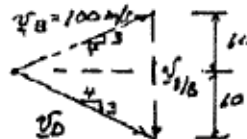
$$v_D = v_B + v_{DB}$$

$$v_{DB} = (10 \text{ in.})\omega_{BDH} \downarrow$$

$$\left[v_D \right] = \left[v_B \right] + [(10)\omega_{BDH} \downarrow]$$

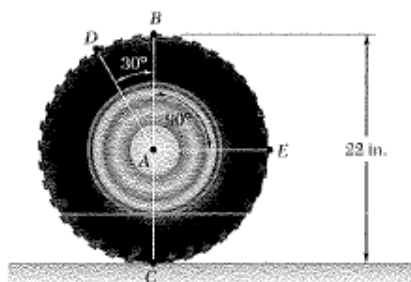
$$v_{DB} = 60 + 60 = 120 \text{ in./s}$$

$$(10 \text{ in.})\omega_{BDH} = 120 \text{ in./s}$$



(a)

$$\omega_{BDH} = 12 \text{ rad/s} \quad \blacktriangleleft$$



PROBLEM 15.70

An automobile travels to the right at a constant speed of 48 mi/h. If the diameter of a wheel is 22 in., determine the velocities of Points *B*, *C*, *D*, and *E* on the rim of the wheel.

SOLUTION

$$v_A = 48 \text{ mi/h} = 70.4 \text{ ft/s}$$

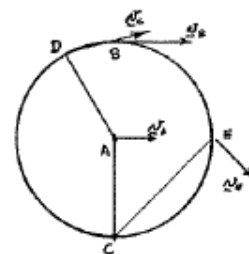
$$v_C = 0 \quad \blacktriangleleft$$

$$d = 22 \text{ in.} \quad r = \frac{d}{2} = 11 \text{ in.} = 0.91667 \text{ ft}$$

$$\omega = \frac{v_A}{r} = \frac{70.4}{0.91667} = 76.8 \text{ rad/s} \quad \curvearrowright$$

$$\begin{aligned} v_{B/A} &= v_{D/A} = v_{E/A} = r\omega \\ &= (0.91667)(76.8) = 70.4 \text{ ft/s} \end{aligned}$$

$$v_B = v_A + v_{B/A} = [70.4 \text{ ft/s} \rightarrow] + [70.4 \text{ ft/s} \rightarrow]$$



$$v_B = 140.8 \text{ ft/s} \rightarrow \quad \blacktriangleleft$$

$$v_D = v_A + v_{D/A} = [70.4 \text{ ft/s} \rightarrow] + [70.4 \text{ ft/s} \angle 30^\circ]$$

$$v_D = 136.0 \text{ ft/s} \angle 15.0^\circ \quad \blacktriangleleft$$

$$v_E = v_A + v_{E/A} = [70.4 \text{ ft/s} \rightarrow] + [70.4 \text{ ft/s} \downarrow]$$

$$v_E = 99.6 \text{ ft/s} \angle 45.0^\circ \quad \blacktriangleleft$$